

Kenny Valent Winalda Sembiring

AI / Machine Learning Engineer · LLM Infrastructure · Local-first AI

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EDUCATION

Universitas Multimedia Nusantara (UMN)

2024 – Est. Aug 2028

S1 Informatika

- Relevant coursework: Machine Learning, Computer Networking, Object-Oriented Programming (Kotlin)

EXPERIENCE

Asset Management Division — Kelompok Studi Pasar Modal

2025 – Present

UMN Student Organization · Tangerang

- Analyzed Indonesian-listed equities (TUGU) for Stock Quest 2026 competition; built full three-statement financial model for TUGU with 2026–2028 projections
- Developed valuation frameworks incorporating PSAK 117 / IFRS 17 adoption impacts on insurance sector reporting
- Produced investment memos and pitch materials for peer review and competition submission

PROJECTS

FinRAG-ID — Indonesian Financial Document Q&A

2025

github.com/KennyUMN/finrag-id

- Built a production-style local-first RAG pipeline on LlamaIndex + Qdrant + Ollama; targets Indonesian financial documents (laporan tahunan, prospektus)
- Implemented hybrid retrieval (BM25 with Sastrawi Indonesian stemmer + BGE-M3 dense embeddings) fused via Reciprocal Rank Fusion for superior recall on numeric financial data
- Runs fully offline via Ollama (Llama 3.1 8B Q5_K_M GGUF quantized); OpenRouter as automatic cloud fallback — stack mirrors real Local AI deployment requirements
- Evaluated with RAGAS metrics (faithfulness, answer relevancy, context precision) + per-query latency tracking; containerized via Docker Compose for one-command deployment
- FastAPI backend with /query, /upload, /health endpoints; Streamlit UI with source citation display

PPE Detection System — YOLOv9c Semi-Supervised Learning

2025 – 2026

IF441 Group Project · Kaggle: kennyvws

- Trained YOLOv9c on a merged Kaggle PPE dataset; implemented a two-phase SSL pipeline: supervised baseline → pseudo-label generation → self-training
- Designed a Compliance State Machine using IoU-based Person-to-PPE mapping for structured safety event detection
- Debugged training pipeline issues (CLI vs Python API), managed multi-phase model checkpointing on Kaggle

Self-Hosted AI Homeserver

2026 – Present

Personal Infrastructure Project · Ubuntu 24.04 / CasaOS

- Running Ubuntu 24.04 + CasaOS with Dockerized services (Ollama, OpenClaw AI gateway, Tailscale VPN overlay)
- Configured OpenClaw to route Telegram/WhatsApp requests through OpenRouter (Llama, Gemini, Claude) with JSON-level config management via SSH
- Practical exposure to local LLM serving, container networking, bind-mount management, and API key rotation

Health Insurance Claims Forecasting — DSC MCF ITB

Feb 2025

- Reduced MAPE from 14.13% to 6.46% using multiplicative seasonal decomposition to preserve monthly seasonality patterns
- Outperformed flat ensemble baseline by explicitly modeling claim volume cycles from 2024 historical data

TECHNICAL SKILLS

AI / LLM

Ollama, vLLM (familiar), LlamaIndex, LangChain (familiar), RAGAS, GGUF quantization, OpenRouter API, HuggingFace

ML / Data	YOLOv9c, Scikit-learn, Pandas, NumPy, Tableau, semi-supervised learning, seasonal decomposition
Infrastructure	Docker, Docker Compose, Ubuntu Server, CasaOS, Tailscale, SSH, Qdrant, ChromaDB
Languages	Python (primary), Kotlin, SQL, Bash
Other	FastAPI, Streamlit, Git/GitHub, REST API integration

COMPETITIONS & ACTIVITIES

- Stock Quest 2026 — Equity Research Competition (Indonesian capital markets, multi-sector analysis)
- DSC MCF ITB 2025 — Data Science Competition (health insurance claims forecasting, top MAPE improvement)
- Dinamic 2025 — Equity Research Competition (Indonesian capital markets, multi-sector analysis)